



Spatial AI in FRC Robotics

Team 4607 – Innovating with 3D Vision



What is Spatial AI?

Spatial AI combines computer vision and depth sensing so robots can understand the world in 3D. Unlike traditional vision, which only provides a 2D location in an image, Spatial AI tells the robot where objects are relative to the robot camera.



How does it work?

- Stereo cameras – Provide depth perception (need feature-rich backgrounds for accuracy).
- Object detection model (e.g., YOLO) – Finds objects in the image (pixel coordinates).
- Fusion – Merges detection + depth to give the robot precise 3D coordinates of objects.



Why does Spatial AI matter to Team 4607?

- Pushes the boundaries of our computer vision capabilities.
- Provides a competitive advantage in matches.
- Unlocks on-the-fly path planning, enabling fully programmatic autonomous routines.



Initial Implementation Plan

Objective: Optimize cycle times with “single-button” object retrieval.

Simplification:

- Detect only game pieces.
- Act on Spatial AI data only when the robot is stationary.

Example (Reefscape):

While the driver/robot places coral, Spatial AI identifies the next game piece and plans retrieval.

After placement, the robot can immediately fetch the next piece, turning a serial process into a parallel process.



Key Impact

- Faster cycles = more points
- Smarter robot behavior
- Stronger technical foundation for future autonomous systems